

What literature is most relevant to interaction design, user interface strategy, human-in-the-loop configuration, and user experience in web-based parametric prosthetic customization platforms that combine anthropometric input, 3D preview, AI-assisted parameter suggestion, and clinician or designer oversight?

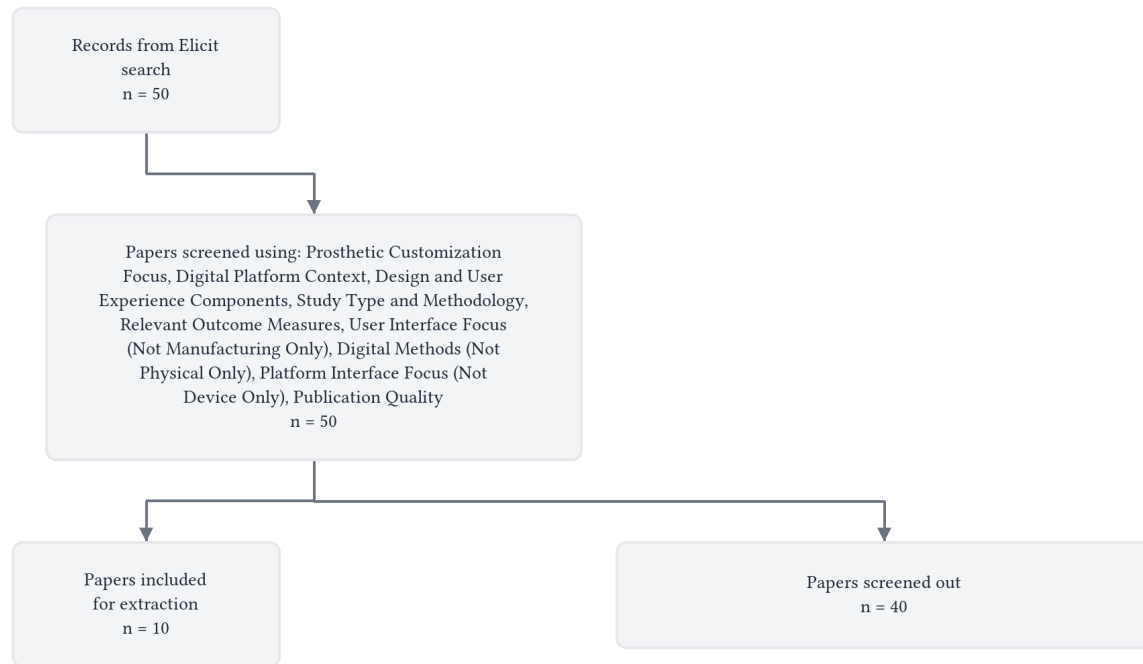
The most relevant literature for web-based parametric prosthetic customization platforms includes Peixoto et al. (2025) as the only web-based implementation addressing interface design and user experience, Colombo et al. (2015) for 3D preview and anthropometric input integration, Abbas Alili et al. (2023) for hierarchical human-in-the-loop configuration with AI-assisted parameter tuning, and Quintero et al. (2018) and Bai et al. (2024) for clinician oversight mechanisms, though no single study addresses all features in a web-based parametric context.

Abstract

The literature on prosthetic customization reveals diverse approaches to interaction design and human-in-the-loop configuration, though direct evidence for web-based parametric platforms remains limited. Only one study explicitly implemented a web-based platform for upper limb aesthetic customization, achieving high usability scores (SUS: 79.5-92.5) through Design Thinking methodology and accessible interface design [1, 1]. Other systems employed desktop [2-4], VR/AR [5, 6], or mobile platforms [7], utilizing direct manipulation interfaces with hand-tracking devices [6], gesture-based input [5], or menu-driven interactions [1, 3, 7]. AI-assisted parameter suggestion was implemented through reinforcement learning for control parameter tuning [2, 4] and knowledge-based rules for design tasks [3, 6], with success depending on training data availability and task structure. Anthropometric input ranged from manual measurement entry and body scanning [6] to real-time biomechanical sensors [4], while 3D preview capabilities were present in geometric customization systems [1, 3, 6] but absent from control parameter tuning applications [2, 4, 8].

Human-in-the-loop integration varied from continuous co-adaptation during gait [4, 5] to discrete clinician oversight [8], with hierarchical approaches combining user preference input, automated optimization, and professional oversight [2]. Clinician oversight mechanisms included direct configuration interfaces reducing tuning time from hours to under 10 minutes [8] and tiered parameter access restricting critical settings to professionals [7]. Usability outcomes differed substantially: simplified interfaces for end-users achieved excellent usability [1], while systems for professional prosthetists exhibited steeper learning curves requiring modifications for non-computer-skilled users [3, 6]. The evidence suggests web-based parametric platforms are feasible for aesthetic and geometric customization tasks, though computationally intensive biomechanical optimization may require hybrid architectures, with optimal AI assistance strategies, anthropometric input methods, and human-in-the-loop integration depending on domain-specific task requirements and user technical backgrounds.

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "What literature is most relevant to interaction design, user interface strategy, human-in-the-loop configuration, and user experience in web-based parametric prosthetic customization platforms that combine anthropometric input, 3D preview, AI-assisted parameter suggestion, and clinician or designer oversight?"

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Prosthetic Customization Focus:** Does this study focus on prosthetic customization or design platforms?
- **Digital Platform Context:** Does this research involve web-based or digital platforms for medical device customization?

- **Design and User Experience Components:** Does this study examine at least one of the following: interaction design, user interface design, human-in-the-loop systems, or user experience in healthcare/assistive technology contexts?
- **Study Type and Methodology:** Is this an empirical study (quantitative, qualitative, or mixed methods), systematic review, meta-analysis, or design case study?
- **Relevant Outcome Measures:** Does this study evaluate usability, user satisfaction, design effectiveness, or clinical outcomes of digital prosthetic design tools?
- **User Interface Focus (Not Manufacturing Only):** Does this study include user interface or design interaction components (i.e., is it NOT focused solely on prosthetic manufacturing processes without interface elements)?
- **Digital Methods (Not Physical Only):** Does this research involve digital or technological interface elements (i.e., is it NOT focused on purely physical prosthetic fitting methods without technological components)?
- **Platform Interface Focus (Not Device Only):** Does this study address the design or customization platform interface (i.e., does it NOT examine prosthetic devices themselves without addressing the platform interface)?
- **Publication Quality:** Is this a peer-reviewed publication (i.e., is it NOT a conference abstract, editorial, opinion piece, or non-peer-reviewed publication)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **System Type:**

Extract the type of prosthetic customization system being studied, including:

- Platform type (web-based, desktop, mobile, VR/AR, etc.)
- Customization approach (parametric, direct modeling, template-based, etc.)
- Target prosthetic type (upper limb, lower limb, specific components)
- Whether it's a design tool, fitting system, or end-user platform
- Development stage (concept, prototype, commercial system)

- **Interface Design Approaches:**

Extract all interaction design and user interface strategies employed in the prosthetic customization system, including:

- Input methods (hand tracking, touch, mouse, voice, gesture, etc.)
- Interaction paradigms (natural user interfaces, direct manipulation, menu-driven, etc.)
- Interface design principles or frameworks applied
- Visualization approaches (2D, 3D, immersive, etc.)
- Navigation and workflow design
- Accessibility considerations for users with disabilities

- **Human-in-the-Loop Configuration:**

Extract how humans (users, clinicians, designers) are integrated into the customization process, including:

- Types of human oversight or input required
- Decision points where human judgment is needed

- Automation vs. manual control balance
- Collaborative features between different user types
- Feedback loops and iterative refinement processes
- Human-AI cooperation or co-adaptive processes
- User control mechanisms and preference incorporation

- **Key Technology Features:**

Identify which of these specific features are present in the system:

- Anthropometric data input (measurement capture, body scanning, manual entry)
- 3D preview/visualization capabilities
- AI-assisted parameter suggestion or automated recommendations
- Clinician or designer oversight mechanisms
- Real-time feedback systems
- Knowledge-based design assistance
- Note any novel or alternative approaches to these features

- **User Experience Findings:**

Extract user experience insights, challenges, and outcomes related to prosthetic customization platforms, including:

- Usability testing results and user satisfaction measures
- User acceptance and adoption factors
- Learning curves and ease of use assessments
- User preferences and requirements identified
- Barriers to effective use
- User empowerment and engagement aspects
- Embodiment and acceptance of prosthetic devices
- Quality of life or functional outcome impacts

- **Design Principles:**

Extract design guidelines, principles, or best practices identified for prosthetic customization systems, including:

- User-centered design approaches
- Human-machine integration strategies
- Personalization and customization methodologies
- Interface design recommendations
- Workflow optimization insights
- Error prevention and recovery strategies
- Accessibility and inclusion principles
- Technology acceptance factors

- **Validation Methods:**

Extract how the system or approach was evaluated, including:

- Study participants (clinicians, amputees, able-bodied users, etc.)
- Evaluation methods (user testing, expert review, comparative studies, etc.)

- Metrics used to assess success
- Comparison approaches or baselines
- Study limitations and scope
- Evidence quality and strength of findings

- **Platform Relevance:**

Assess and extract the direct relevance of this study to web-based parametric prosthetic customization platforms, including:

- Applicability of findings to web-based systems specifically
- Transferability to parametric customization approaches
- Relevance to the target feature set (anthropometric input, 3D preview, AI assistance, clinician oversight)
- Gaps or limitations in addressing the research question
- Novel insights that advance understanding of such platforms
- Whether findings contradict or support other approaches

Results

Characteristics of Included Studies

Study	Full text retrieved?	Platform Type	Target Prosthetic Type	Customization Approach	System Type	Development Stage
Abbas Alili et al., 2023 [2]	Yes	Desktop/mobile (not specified) [2]	Lower limb (robotic knee) [2]	Parametric (reinforcement learning) [2]	Design/fitting tool [2]	Prototype [2]
Ricardo Garcia-Rosas et al., 2019 [5]	Yes	VR/AR (Oculus Rift) [5]	Upper limb [5]	Parametric (kinematic synergies) [5]	Fitting system [5]	Prototype [5]
Sara Peixoto et al., 2025 [1]	Yes	Web-based [1]	Upper limb [1]	User-driven personalization [1]	End-user platform [1]	Prototype [1]
Stella Gabbiadini, 2011 [3]	No	Desktop [3]	Lower limb [3]	Parametric [3]	Design tool [3]	Prototype [3]
Bretta L. Fylstra et al., 2022 [4]	Yes	Desktop/computer based [4]	Lower limb (powered knee) [4]	Adaptive/reinforcement learning [4]	Design/fitting tool [4]	Prototype/experimental [4]
David Quintero et al., 2018 [8]	No	Desktop/clinical software (not specified) [8]	Lower limb (powered knee-ankle) [8]	Parametric [8]	Fitting system [8]	Prototype [8]

Study	Full text retrieved?	Platform Type	Target Prosthetic Type	Customization Approach	System Type	Development Stage
Xiaolu Bai et al., 2024 [7]	Yes	Mobile or desktop [7]	Lower limb [7]	Parametric [7]	End-user platform [7]	Prototype and commercial [7]
Andrea Marinelli et al., 2022 [9]	Yes	Not specified [9]	Upper limb [9]	Not specified [9]	Not specified [9]	Not specified [9]
G. Colombo et al., 2015 [6]	Yes	Desktop/VR/AR [6]	Lower limb (socket) [6]	Parametric/direct modeling [6]	Design tool [6]	Prototype [6]
P. Beckerle, 2014 [10]	No	Not specified [10]	Lower limb (powered knee) [10]	Human-machine-centered design framework [10]	Design/fitting tool [10]	Not specified [10]

The included studies represent a diverse range of prosthetic customization approaches across both upper and lower limb systems. The majority focused on lower limb prosthetics (7 studies) [2–4, 6–8, 10], with three addressing upper limb applications [1, 5, 9]. Platform types varied widely, with only one study explicitly implementing a web-based platform [1], while others utilized desktop [3, 4, 6], VR/AR [5, 6], or mobile interfaces [7]. Most systems employed parametric customization approaches [2, 3, 5–8], though specific implementation strategies differed substantially. The majority of systems were in the prototype stage [1–6, 8], with only one study reporting both prototype and commercial implementations [7].

Thematic Analysis

Interface Design Strategies and Interaction Paradigms

The studies revealed diverse approaches to user interface design for prosthetic customization. Direct manipulation interfaces were prominent, with Abbas Alili et al. implementing a User-Controlled Interface where users modified control points via remote controller [2]. Similarly, Colombo et al. developed a Natural User Interface using Leap Motion hand-tracking devices to emulate traditional manual design tasks [6], while Garcia-Rosas et al. employed gesture-based input through Oculus Touch controllers for upper limb motion tracking in a VR environment [5].

Menu-driven interaction paradigms were also common. Gabbiadini's framework utilized virtual assistants providing structured procedures and suggestions [3], while Bai et al. described user-controlled interfaces allowing wearers to adjust settings through menu-driven interactions [7]. Peixoto et al.'s web-based platform implemented direct manipulation through buttons and navigation hierarchies [1].

Visualization approaches varied by platform type. Immersive 3D visualization was implemented in Garcia-Rosas et al.'s VR environment using Oculus Rift [5] and Colombo et al.'s desktop/VR system [6]. Gabbiadini utilized 3D CAD visualization [3], while Peixoto et al. provided real-time 2D visualization with 360-degree view capabilities [1]. Fylstra et al. employed screen-based visual feedback displaying real-time ground reaction force signals [4].

Accessibility considerations were explicitly addressed in several studies. Peixoto et al. incorporated digital accessibility tools, chatbots, large buttons for physical disabilities, and simple language [1]. Bai et al. recommended customization options for font, color, and text size, along with additional sensory channels for individuals with visual impairments [7]. Colombo et al. designed their interface to minimize fatigue and enable intuitive interaction for users without IT skills [6].

Human-in-the-Loop Configuration and Co-Adaptation

The integration of human oversight varied substantially across customization approaches. User control mechanisms ranged from selecting preferred kinematics [2] to modifying control points [2] and adjusting stance time through visual feedback [4]. Garcia-Rosas et al. demonstrated concurrent human-AI adaptation where the system continuously adjusted synergy settings based on user performance [5], while Fylstra et al. explored how users controlled gait timing through visual feedback that cooperated with automatic prosthesis tuning [4].

The balance between automation and manual control was explicitly managed in multiple systems. Abbas Alili et al. allowed human input through a User-Controlled Interface which was then adjusted by a reinforcement learning algorithm [2]. Bai et al. limited the parameters accessible to wearers while reserving more complex adjustments for prosthetists [7]. Colombo et al. provided both automatic and interactive modes for shape manipulation [6].

Feedback loops and iterative refinement were central to several approaches. Abbas Alili et al. used qualitative surveys to assess interface usability [2], Garcia-Rosas et al. adapted synergy settings iteratively based on user performance [5], and Peixoto et al. incorporated confirmation messages for critical actions and chatbot support [1]. Fylstra et al. implemented visual feedback that users employed to adjust stance time repeatedly during tuning phases [4].

Collaborative features between different user types were noted in several systems. Clinicians could integrate user preferences into the tuning process [2], while prosthetists supervised production to ensure precision [1]. Quintero et al.'s system was specifically designed for certified prosthetists to configure powered prostheses [8]. Colombo et al.'s virtual tools enabled collaboration between designers and clinicians [6].

Technology Feature Implementation

Anthropometric Data Input

Anthropometric data capture was implemented through various methods. Colombo et al. utilized measurement capture, body scanning, and manual entry to create virtual models based on patient data [6]. Gabbiadini employed digital models of the patient's body [3]. Fylstra et al. incorporated real-time ground reaction force signals as a form of anthropometric input [4]. However, several studies did not explicitly mention anthropometric input mechanisms [1, 9, 10].

3D Preview and Visualization

Three-dimensional preview capabilities were present in multiple systems. Peixoto et al.'s web platform provided real-time visualization with 360-degree view functionality [1]. Colombo et al. implemented comprehensive 3D visualization capabilities for virtual socket modeling [6]. Gabbiadini's Socket Modelling Assistant enabled 3D virtual modeling of prosthetic components [3]. Several studies did not mention 3D preview features [4, 7, 8, 10].

AI-Assisted Parameter Suggestion

AI assistance was implemented through diverse mechanisms. Abbas Alili et al. used reinforcement learning algorithms to adjust parameters based on user input [2]. Garcia-Rosas et al. employed model-guided data-driven

optimization for automated synergy recommendations [5]. Gabbiadini utilized virtual assistants providing automated procedures and suggestions [3]. Fylstra et al. incorporated AI-assisted suggestions through custom code and MATLAB-based parameter adjustment [4]. Bai et al. described human-in-the-loop optimization where computers adjusted parameters based on wearer responses [7].

Clinician and Designer Oversight

Clinician oversight mechanisms varied across systems. Quintero et al. developed an intuitive clinician control interface specifically for certified prosthetists to configure prostheses [8]. Gabbiadini's framework implicitly supported clinician oversight to guarantee high-quality products [3]. Bai et al. allowed some settings to be accessible only by prosthetists while wearers could adjust essential parameters [7]. Marinelli et al. emphasized the involvement of clinicians and designers in the development process [9].

Real-Time Feedback Systems

Real-time feedback was a critical feature in several implementations. Abbas Alili et al. enabled users to modify control points with immediate feedback [2]. Garcia-Rosas et al. used real-time performance data to adjust synergy settings [5]. Fylstra et al. implemented visual feedback systems displaying continuous stance time information [4]. Quintero et al. provided real-time feedback through continuous gait phase measurement [8]. Bai et al. described intelligent prostheses using embedded sensors for real-time monitoring [7]. Colombo et al. incorporated confidence percentages for detected gestures as real-time feedback [6].

Knowledge-Based Design Assistance

Knowledge-based systems were implemented in multiple studies. Gabbiadini developed a central knowledge-based framework providing rules and procedures for design tasks [3]. Quintero et al. used knowledge-based design through virtual constraint control schemes [8]. Colombo et al.'s Socket Modelling Assistant emulated traditional manufacturing processes using knowledge-based rules [6].

User Experience and Usability Outcomes

Usability testing revealed generally positive outcomes across studies. Peixoto et al. reported high System Usability Scale scores (92.5, 79.5, 86.25) indicating excellent usability [1]. Abbas Alili et al. found their User-Controlled Interface easy to use for 7 out of 10 participants [2]. Quintero et al. demonstrated that clinicians could tune powered prostheses in under 10 minutes compared to hours with traditional methods [8].

User acceptance factors were influenced by multiple elements. Users could accurately and consistently identify their preferred prosthetic control profiles [2]. Garcia-Rosas et al.'s algorithm adapted to individual motor preferences and learning characteristics, achieving performance steady-state under 50 iterations on average [5]. Peixoto et al.'s platform was found intuitive, aesthetically pleasing, and accessible even for users with low digital literacy [1].

Learning curves varied by system complexity. Peixoto et al.'s onboarding process was praised for clarity and fluidity with a low learning curve [1]. Garcia-Rosas et al. achieved uniform convergence across subjects regardless of individual motor learning characteristics [5]. Colombo et al. observed a steep learning curve for beginners who required repeated attempts to execute gestures correctly, while experts were more adept [6].

Barriers to effective use were identified in several studies. Abbas Alili et al. noted participants required additional explanation of knee kinematics [2]. Gabbiadini found modifications were needed to make tools easier to use for non-computer-skilled users [3]. Fylstra et al. reported that visual feedback effects were not sustained after removal, indicating challenges in long-term retention [4]. Colombo et al. identified initial issues with ergonomics and fatigue during long interactions [6].

User empowerment and engagement were enhanced through various mechanisms. Abbas Alili et al. enabled participants to physiologically differentiate between profiles and recognize preferred settings with moderate accuracy [2]. Bai et al. emphasized intuitive involvement and optimal control as goals for user empowerment [7]. Marinelli et al. stressed the importance of user-centered design and training to enhance engagement [9].

Quality of life and functional outcomes were addressed by several studies. Peixoto et al.'s high usability scores suggested positive impacts on quality of life [1]. Quintero et al. improved amputee gait outcome metrics such as gait symmetry [8]. Bai et al. linked satisfaction and embodiment to overall quality of life impacts [7]. Marinelli et al. aimed to improve prosthetic innovation toward human-centered artificial limbs to enhance quality of life [9].

Design Principles and Methodological Approaches

User-centered design emerged as a foundational principle across multiple studies. Abbas Alili et al. emphasized personalized assistance for individual prosthesis users [2]. Peixoto et al. employed Design Thinking methodology centered on user needs [1]. Marinelli et al. highlighted patient-tailored design processes as crucial for device effectiveness [9]. Beckerle developed a human-machine-centered design approach equally considering technical and human factors [10].

Personalization and customization methodologies varied in implementation sophistication. Garcia-Rosas et al. used grey-box models to capture individual motor preference and adaptation [5]. Abbas Alili et al. employed reinforcement learning to tune parameters based on user preference [2]. Peixoto et al. focused on personalization to reduce social stigma and promote inclusion [1]. Quintero et al. enabled tailored adjustments to improve patient outcomes [8].

Interface design recommendations included multiple specific strategies. Bai et al. advised avoiding cluttered interfaces and providing accessible displays for users with impairments [7]. Peixoto et al. prioritized intuitive and efficient navigation with clear visual and functional experiences [1]. Garcia-Rosas et al. recommended algorithms that estimate steady-state performance to adapt to user learning stages [5]. Colombo et al. emphasized ergonomics, posture, and environment considerations with virtual space mapping [6].

Workflow optimization strategies were implemented through various approaches. Abbas Alili et al. demonstrated framework effectiveness and efficiency in tuning parameters while ensuring user safety [2]. Quintero et al. significantly reduced configuration time from hours to under 10 minutes through their clinician control interface [8]. Garcia-Rosas et al. achieved uniform convergence performance reducing the need for manual tuning [5]. Peixoto et al. utilized wireframes, wireflows, and design systems for workflow optimization [1].

Error prevention and recovery strategies included multiple mechanisms. Abbas Alili et al. ensured user safety during parameter tuning [2]. Fylstra et al. found human control did not negatively impact prosthesis tuning [4]. Bai et al. recommended implementing mistake prevention, detection, and fail-safe mechanisms [7]. Colombo et al. used feedback mechanisms like icons and confidence percentages to prevent errors [6].

Accessibility and inclusion principles were explicitly addressed by several frameworks. Peixoto et al. incorporated digital accessibility tools with focus on users with upper limb amputations [1]. Bai et al. recommended considering needs of users with visual, hearing, or cognitive impairments [7]. Colombo et al. tested their system with volunteers to ensure accessibility and minimize fatigue for users without IT skills [6]. Marinelli et al. advocated for human-centered artificial limbs designed for universal applicability [9].

Validation Approaches and Evidence Quality

Study participant composition varied substantially across research. Abbas Alili et al. recruited ten participants including five non-disabled individuals and five with unilateral above-knee amputation [2]. Garcia-Rosas et al. tested with twelve able-bodied subjects in a VR environment [5]. Peixoto et al. evaluated their platform with users across diverse digital literacy levels and physical abilities [1]. Gabbiadini tested with clinicians and technicians from orthopaedic laboratories [3]. Fylstra et al. included seven individuals without amputation and four with transfemoral amputation [4].

Evaluation methodologies employed multiple complementary approaches. User testing was the most common method [1–6, 8], often combined with comparative studies [2–4]. Peixoto et al. utilized the System Usability Scale post-test questionnaire [1]. Fylstra et al. employed statistical analysis using Chi-square tests and ANOVAs [4]. Bai et al. proposed comprehensive evaluation frameworks using NASA-TLX, EEG, fNIRS, and subjective measures like TAPES and PEQ [7].

Success metrics included both objective and subjective measures. Abbas Alili et al. assessed success rates in meeting targeted knee kinematics and tuning duration [2]. Garcia-Rosas et al. measured algorithm adaptation ability and uniform convergence speed across subjects [5]. Peixoto et al. relied on SUS scores for usability assessment [1]. Quintero et al. evaluated improvement in gait outcome metrics such as gait symmetry [8]. Colombo et al. assessed ergonomics, ease of use, and precision [6].

Comparison baselines varied across studies. Abbas Alili et al. compared user-preferred profiles against normative and opposite profiles [2]. Garcia-Rosas et al. compared their model-guided approach with a previous black-box optimization technique [5]. Fylstra et al. tested different visual feedback conditions (no feedback, self-selected stance time feedback, increased stance time feedback) [4]. Quintero et al. compared clinician tuning time against traditional engineering methods [8].

Study limitations were explicitly acknowledged in several works. Abbas Alili et al. noted short study timeframes and limited sample sizes [2]. Garcia-Rosas et al. acknowledged their study was limited to able-bodied subjects and a specific reaching task [5]. Peixoto et al. recognized restricted generalizability due to limited sample size [1]. Fylstra et al. highlighted the small number of participants with amputation and potential non-transferability from able-bodied to amputee populations [4]. Colombo et al. was limited by small sample size and lack of detailed statistical analysis [6].

Evidence quality varied based on study design rigor. Garcia-Rosas et al. demonstrated strong evidence quality through controlled experimental setup with ethical approval [5]. Fylstra et al. used structured statistical analysis with significance levels set at $\alpha = 0.05$ [4]. Peixoto et al. focused strongly on user-centered design with high SUS scores [1]. Gabbiadini provided qualitative assessment considered preliminary and positive but requiring modifications [3].

Synthesis

The reviewed literature reveals substantial heterogeneity in approaches to prosthetic customization platforms, with varying degrees of direct relevance to web-based parametric systems. Only one study explicitly implemented a web-based platform [1], while the majority developed desktop [2–4, 6], VR/AR [5, 6], or unspecified systems [7–10]. This platform diversity reflects the evolution of prosthetic customization technology rather than contradictory findings—each platform type addresses specific use cases and technical constraints.

The apparent contradiction between highly automated systems [2, 4, 5] and user-driven customization platforms [1] can be explained through context-specific appropriateness. Automated approaches using reinforcement learning

[2] and model-guided optimization [5] were implemented for parametric control tuning requiring real-time biomechanical optimization—contexts where manual parameter adjustment would be impractical. Conversely, aesthetic personalization platforms [1] prioritized user autonomy since preferences for appearance are inherently subjective and not amenable to algorithmic optimization. Both approaches are valid within their respective domains: biomechanical control versus aesthetic customization.

The variability in human-in-the-loop integration strategies—from continuous co-adaptation [5] to discrete clinician oversight [8]—reflects different stages of the prosthetic lifecycle. Continuous co-adaptation during gait [4, 5] addresses real-time functional optimization during prosthesis use, while clinician-mediated configuration [8] targets initial setup and periodic adjustments. These are complementary rather than competing approaches, with Abbas Alili et al. demonstrating that both user preference input and automated tuning can be integrated hierarchically [2].

The sparse implementation of web-based platforms [1] versus desktop systems [2–4, 6] likely reflects technical constraints rather than fundamental limitations. Desktop and VR/AR implementations provided access to high-performance computing for real-time biomechanical simulation [4–6], computational demands potentially challenging for web browsers during these studies' development periods. However, Peixoto et al.'s successful web-based implementation for aesthetic customization [1] demonstrates feasibility for applications not requiring intensive real-time computation, suggesting web platforms may be appropriate for certain customization aspects while desktop systems handle computationally intensive tasks.

The heterogeneity in AI assistance implementation—from reinforcement learning [2, 4] to knowledge-based rules [3, 6, 8]—can be understood through the availability of training data and task structure. Reinforcement learning approaches [2, 4] were applied to gait control where extensive sensor data and clear optimization objectives exist. Knowledge-based systems [3, 6] were implemented for socket design where expert rules could be formalized but sufficient training data for machine learning was unavailable. This suggests AI assistance strategies should be selected based on domain-specific data availability and task characteristics rather than applying uniform approaches.

The variation in anthropometric input methods—from manual measurement entry [6] to real-time sensor data [4]—represents different points on the accuracy-convenience tradeoff spectrum. Real-time ground reaction force signals [4] provided continuous biomechanical data for dynamic control optimization but required specialized equipment. Manual entry and body scanning [6] enabled socket design from static measurements without specialized sensors but lacked dynamic feedback. The optimal approach depends on whether static geometry or dynamic biomechanics drives customization requirements.

Regarding 3D preview implementation, the studies demonstrating visualization capabilities [1, 3, 6] focused on geometric customization tasks (socket design, aesthetic personalization) where spatial understanding is critical. Studies omitting 3D preview [2, 4, 5, 8] addressed parametric control tuning where abstract parameters rather than geometry required adjustment. This suggests 3D preview is essential for geometric customization but potentially less critical for pure control parameter tuning, though integration could enhance user understanding.

The divergent usability outcomes—from excellent scores [1] to identified barriers [2, 3, 6]—can be reconciled through examining target user populations and task complexity. Peixoto et al.'s high usability scores [1] were achieved with a web platform for aesthetic customization targeting end-users through simplified interactions. Lower usability and steeper learning curves [6] occurred in systems for professional prosthetists performing complex geometric modeling requiring technical expertise. Gabbiadini's finding that modifications were needed for non-computer-skilled users [3] supports the interpretation that task complexity and user technical background drive usability outcomes more than platform type per se.

The variation in clinician oversight mechanisms—from direct configuration interfaces [8] to parameter restriction schemes [7]—reflects different safety-autonomy balancing strategies. Quintero et al.'s full clinician control [8] pri-

oritized safety and expertise for powered prosthesis configuration in clinical settings. Bai et al.'s tiered parameter access [7] enabled user autonomy for routine adjustments while restricting critical parameters to professionals. Both approaches are appropriate within specific regulatory and liability contexts, with the optimal balance depending on risk assessment and user competency levels.

For web-based parametric prosthetic customization platforms combining anthropometric input, 3D preview, AI assistance, and clinician oversight, the synthesis suggests several evidence-based recommendations. First, web platforms are feasible for aesthetic customization [1] but may require hybrid architectures for computationally intensive biomechanical optimization. Second, AI assistance should employ reinforcement learning for control parameter tuning [2, 4] when training data exists, and knowledge-based rules [3, 6] for design tasks where expert knowledge is formalized but data is limited. Third, 3D preview is essential for geometric customization [1, 6] and should be real-time [1] to support iterative user refinement. Fourth, human-in-the-loop integration should be hierarchical [2], combining user preference input, automated optimization, and clinician oversight with parameter access tiered by risk level [7]. Fifth, anthropometric input should match task requirements—static measurements for geometry [6], dynamic sensors for gait control [4]. Finally, usability optimization should account for end-user technical background, with simplified interfaces for consumers [1] and more sophisticated tools for professionals [6].

References

1. Peixoto S, Martins N, Miranda D, et al (2025) The Design Process in the Development of an Online Platform for Personalizing Wearable Prostheses: A Preliminary Approach. *Designs*. <https://doi.org/10.3390/designs9020039>
2. Alili A, Nalam V, Li M, et al (2023) A Novel Framework to Facilitate User Preferred Tuning for a Robotic Knee Prosthesis. *IEEE transactions on neural systems and rehabilitation engineering*. <https://doi.org/10.1109/TNSRE.2023.3236217>
3. Gabbiadini S (2011) Knowledge-based design of lower limb prosthesis
4. Fylstra BL, Lee I-C, Li M, et al (2022) Human-prosthesis cooperation: combining adaptive prosthesis control with visual feedback guided gait. *Journal of NeuroEngineering and Rehabilitation*. <https://doi.org/10.1186/s12984-022-01118-z>
5. Garcia-Rosas R, Tan Y, Oetomo D, et al (2019) Personalized Online Adaptation of Kinematic Synergies for Human-Prosthesis Interfaces. *IEEE Transactions on Cybernetics*. <https://doi.org/10.1109/TCYB.2019.2920376>
6. Colombo G, Facoetti G, Rizzi C, Vitali A (2015) Low Cost Hand-Tracking Devices to Design Customized Medical Devices. *Interacción*. https://doi.org/10.1007/978-3-319-21067-4_36
7. Bai X, Yuan J, Liu M, et al (2024) Human factors considerations of Interaction between wearers and intelligent lower-limb prostheses: a prospective discussion. *Journal of NeuroEngineering and Rehabilitation*. <https://doi.org/10.1186/s12984-024-01475-x>
8. Quintero D, Reznick E, Lambert DJ, et al (2018) Intuitive Clinician Control Interface for a Powered Knee-Ankle Prosthesis: A Case Study. *IEEE Journal of Translational Engineering in Health and Medicine*. <https://doi.org/10.1109/JTEHM.2018.2880199>

9. Marinelli A, Boccardo N, Tessari F, et al (2022) Active upper limb prostheses: a review on current state and upcoming breakthroughs. *Progress in Biomedical Engineering*. <https://doi.org/10.1088/2516-1091/acac57>
10. Beckerle P (2014) Human-Machine-Centered Design and Actuation of Lower Limb Prosthetic Systems